Class14

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Section 1. Differential Expression Analysis

You can download the count data and associated metadata from here: GSE37704_featurecounts.csv and GSE37704_metadata.csv. This is similar to our starting point for the last class where we used DESeq2 for the first time. We will use it again today!

```
library(DESeq2)
```

```
Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics
```

```
Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo
```

```
Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians
```

```
The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

Import metadata and take a peek:

```r
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

Import countdata

```r
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
                length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Hmm… remember that we need the `countData` and `colData` files to match up so we will need to remove that odd first column in `countData` namely `countData$length`.

Q. Complete the code below to remove the troublesome first column from countData

```
countData <- as.matrix(countData[, -1])
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

This looks better but there are lots of zero entries in there so let's get rid of them as we have no data for these.

> Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns). Tip: What will rowSums() of countData return and how could you use it in this context?

```
countData <- countData[rowSums(countData) != 0, ]
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

Running DESeq2

Now lets setup the DESeqDataSet object required for the DESeq() function and then run the DESeq pipeline. This is again similar to our last day's hands-on session.

```r
colData <- read.csv(metaFile, header=TRUE, stringsAsFactors=FALSE)
colData <- as.data.frame(colData)

rownames(colData) <- colData[,1]

colData <- colData[ , -1, drop=FALSE]
```

```r
class(colData)
```

```
[1] "data.frame"
```

```r
dim(colData)
```

```
[1] 6 1
```

```r
rownames(colData)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```r
countData <- as.matrix(countData)

colData <- colData[match(colnames(countData), rownames(colData)), , drop=FALSE]

colData$condition <- factor(colData$condition)

dds <- DESeqDataSetFromMatrix(
  countData = countData,
  colData = colData,
  design = ~ condition
)

dds <- DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates
```

```
mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```r
res <- results(dds)
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.9798      0.5428105 0.5215598   1.040744 2.97994e-01
                      padj
                 <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76549e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
ENSG00000187642 4.03379e-01
```

This gives you: - baseMean - log2FoldChange - lfcSE - stat - pvalue - padj

```r
resOrdered <- res[order(res$padj), ]

head(resOrdered)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns
                 baseMean log2FoldChange     lfcSE      stat    pvalue
                <numeric>      <numeric> <numeric> <numeric> <numeric>
```

```
ENSG00000117519   4483.63       -2.42272 0.0600016  -40.3776         0
ENSG00000183508   2053.88        3.20196 0.0724172   44.2154         0
ENSG00000159176   5692.46       -2.31374 0.0575534  -40.2016         0
ENSG00000116016   4423.95       -1.88802 0.0431680  -43.7366         0
ENSG00000164251   2348.77        3.34451 0.0690718   48.4208         0
ENSG00000124766   2576.65        2.39229 0.0617086   38.7675         0
                     padj
                <numeric>
ENSG00000117519         0
ENSG00000183508         0
ENSG00000159176         0
ENSG00000116016         0
ENSG00000164251         0
ENSG00000124766         0
```

> Q. Call the summary() function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```r
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

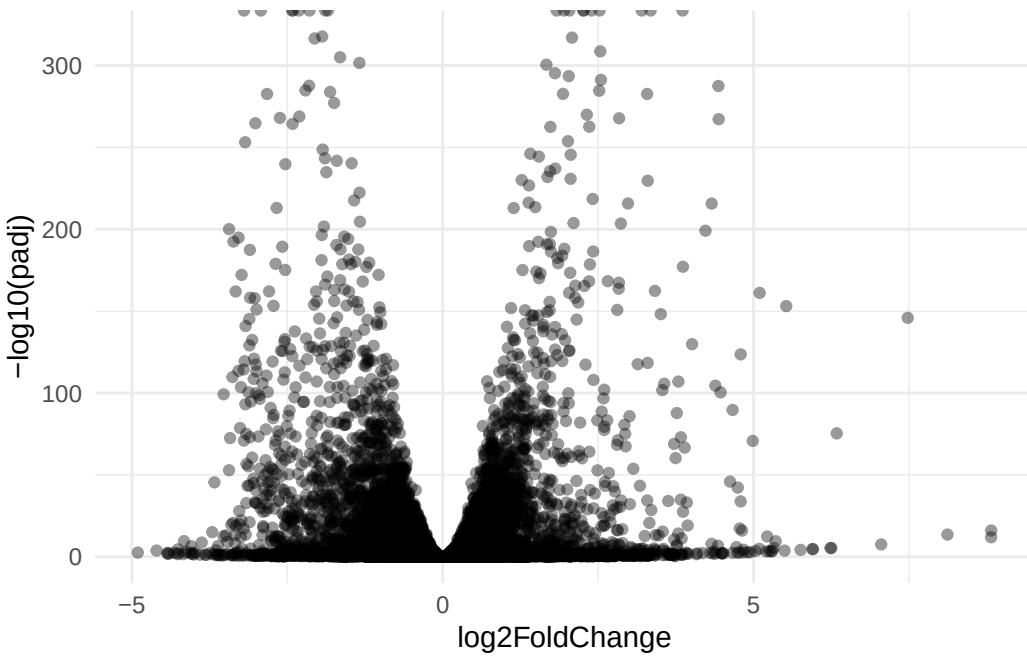
Volcano Plot

Now we will make a volcano plot, a commonly produced visualization from this type of data that we introduced last day. Basically it's a plot of log2 fold change vs -log adjusted p-value.

```r
library(ggplot2)

res_df <- as.data.frame(res)

ggplot(res_df, aes(x=log2FoldChange, y=-log10(padj))) +
```

```
  geom_point(alpha=0.4) +
  theme_minimal()
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



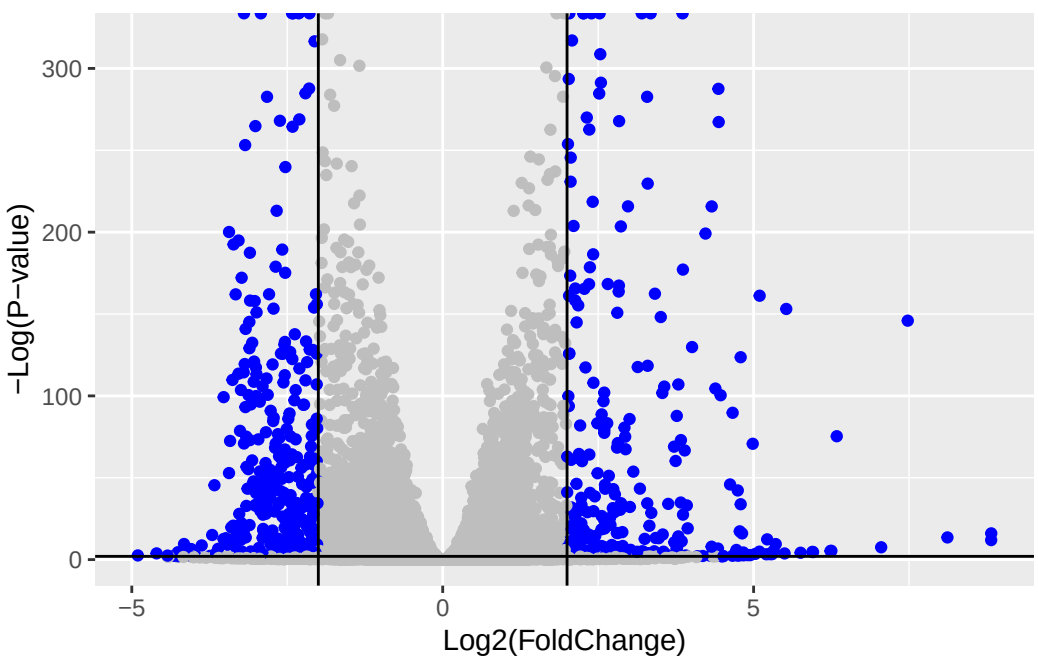
Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
res_df <- as.data.frame(res)
res_df <- na.omit(res_df)
res_df$significant <- ifelse(
  res_df$padj < 0.05 & abs(res_df$log2FoldChange) > 1,
  "Significant",
  "Not Significant"
)
```

```
mycols <- rep("gray", nrow(res))

mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"
```

```
mycols[ res$padj > 0.01 ] <- "gray"

ggplot(res) +
  aes(x = log2FoldChange,
      y = -log10(padj)) +
  geom_point(color = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log10(0.01))
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Adding gene annotation

Since we mapped and counted against the Ensembl annotation, our results only have information about Ensembl gene IDs. However, our pathway analysis downstream will use KEGG pathways, and genes in KEGG pathways are annotated with Entrez gene IDs. So lets add them as we did the last day.

Q. Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
if (!requireNamespace("BiocManager", quietly = TRUE))
    install.packages("BiocManager")

BiocManager::install("AnnotationDbi")
```

```
Bioconductor version 3.22 (BiocManager 1.30.27), R 4.5.2 (2025-10-31)

Warning: package(s) not installed when version(s) same as or greater than current; use
  `force = TRUE` to re-install: 'AnnotationDbi'
```

```
BiocManager::install("org.Hs.eg.db")
```

```
Bioconductor version 3.22 (BiocManager 1.30.27), R 4.5.2 (2025-10-31)

Warning: package(s) not installed when version(s) same as or greater than current; use
  `force = TRUE` to re-install: 'org.Hs.eg.db'
```

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol = mapIds(org.Hs.eg.db,
                    keys = rownames(res),
                    keytype = "ENSEMBL",
                    column = "SYMBOL",
                    multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$entrez = mapIds(org.Hs.eg.db,
                    keys = rownames(res),
                    keytype = "ENSEMBL",
                    column = "ENTREZID",
                    multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$name = mapIds(org.Hs.eg.db,
                  keys = rownames(res),
                  keytype = "ENSEMBL",
                  column = "GENENAME",
                  multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                   baseMean log2FoldChange     lfcSE       stat      pvalue
                  <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.913579      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.229650      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.188076     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.637938      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.255123      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.979750      0.5428105 0.5215598   1.040744 2.97994e-01
ENSG00000188290  108.922128      2.0570638 0.1969053  10.446970 1.51282e-25
ENSG00000187608  350.716868      0.2573837 0.1027266   2.505522 1.22271e-02
ENSG00000188157 9128.439422      0.3899088 0.0467163   8.346304 7.04321e-17
ENSG00000237330    0.158192      0.7859552 4.0804729   0.192614 8.47261e-01
                     padj      symbol      entrez                   name
                  <numeric> <character> <character>            <character>
ENSG00000279457 6.86555e-01          NA          NA                     NA
ENSG00000187634 5.15718e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.76549e-35       NOC2L       26155 NOC2 like nucleolar ..
ENSG00000187961 1.13413e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.19031e-01     PLEKHN1       84069 pleckstrin homology ..
ENSG00000187642 4.03379e-01       PERM1       84808 PPARGC1 and ESRR ind..
ENSG00000188290 1.30538e-24        HES4       57801 hes family bHLH tran..
ENSG00000187608 2.37452e-02       ISG15        9636 ISG15 ubiquitin like..
ENSG00000188157 4.21963e-16        AGRN      375790                  agrin
ENSG00000237330          NA      RNF223      401934 ring finger protein ..
```

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```r
resOrdered <- res[order(res$padj), ]
resOrdered <- res[order(res$padj, na.last = NA), ]
```

```r
write.csv(resOrdered,
          file = "deseq2_results_ordered_by_padj.csv")
```

Section 2. Pathway Analysis

Here we are going to use the gage package for pathway analysis. Once we have a list of enriched pathways, we're going to use the pathview package to draw pathway diagrams, shading the molecules in the pathway by their degree of up/down-regulation.

```r
library(pathview)
```

```
##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

```r
library(gage)
```

```r
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"
```

The main gage() function requires a named vector of fold changes, where the names of the values are the Entrez gene IDs.

```r
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez

foldchanges <- foldchanges[!is.na(names(foldchanges))]
```

```r
foldchanges <- foldchanges[!is.na(foldchanges)]

head(foldchanges)
```

```
     148398       26155      339451       84069       84808       57801 
 0.42645712 -0.69272046  0.72975561  0.04057653  0.54281049  2.05706383 
```

```r
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

See help on the gage function with ?gage. Specifically, you might want to try changing the value of same.dir. This value determines whether to test for changes in a gene set toward a single direction (all genes up or down regulated) or changes towards both directions simultaneously (i.e. any genes in the pathway dysregulated). Here, we're using the default same.dir=TRUE, which will give us separate lists for pathways that are upregulated versus pathways that are down-regulated.

Now lets look at the object returned from gage().

```r
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

It is a list with three elements, "greater", "less" and "stats".

You can also see this in your Environment panel/tab window of RStudio or use the R command str(keggres).

Like any list we can use the dollar syntax to access a named element, e.g. `head(keggres$greater)` and `head(keggres$less)`.

Lets look at the first few down (less) pathway results:

```r
head(keggres$less)
```

```
                                        p.geomean stat.mean        p.val
hsa04110 Cell cycle                   9.366328e-06 -4.369134 9.366328e-06
hsa03030 DNA replication              9.592718e-05 -3.946505 9.592718e-05
hsa03013 RNA transport                1.305049e-03 -3.045177 1.305049e-03
hsa03440 Homologous recombination     3.106306e-03 -2.848189 3.106306e-03
hsa04114 Oocyte meiosis               3.888649e-03 -2.688749 3.888649e-03
```

```
hsa00010 Glycolysis / Gluconeogenesis 9.125398e-03 -2.398384 9.125398e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                   0.001507979      121 9.366328e-06
hsa03030 DNA replication              0.007722138       36 9.592718e-05
hsa03013 RNA transport                0.070037607      144 1.305049e-03
hsa03440 Homologous recombination     0.125028834       28 3.106306e-03
hsa04114 Oocyte meiosis               0.125214484      102 3.888649e-03
hsa00010 Glycolysis / Gluconeogenesis 0.213990959       53 9.125398e-03
```

Each keggres*lessandkeggres*greater object is data matrix with gene sets as rows sorted by p-value.

The top "less/down" pathways is "Cell cycle" with the KEGG pathway identifier hsa04110.

Now, let's try out the pathview() function from the pathview package to make a pathway plot with our RNA-Seq expression results shown in color. To begin with lets manually supply a pathway.id (namely the first part of the "hsa04110 Cell cycle") that we could see from the print out above.

```r
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14
```

```
Info: Writing image file hsa04110.pathview.png
```

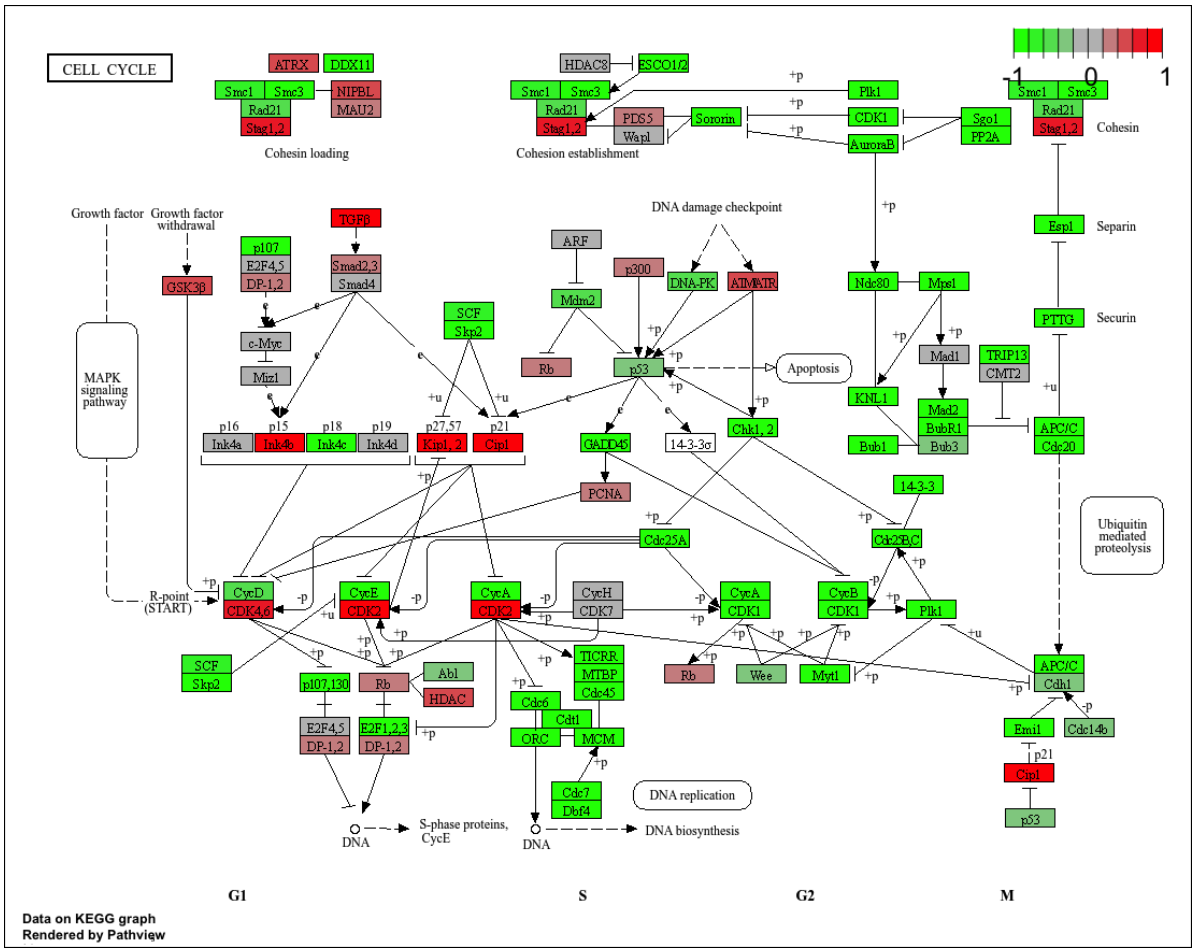


Figure 1: My Plot

Note how many of the genes in this pathway are perturbed (i.e. colored) in our results.

You can play with the other input arguments to pathview() to change the display in various ways including generating a PDF graph. For example:

```
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

```
'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

     [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

```
Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa04110.pathview.pdf
```

Now, let's process our results a bit more to automagicaly pull out the top 5 upregulated pathways, then further process that just to get the pathway IDs needed by the pathview() function. We'll use these KEGG pathway IDs for pathview plotting below.

```
keggrespathways <- rownames(keggres$greater)[1:5]

keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

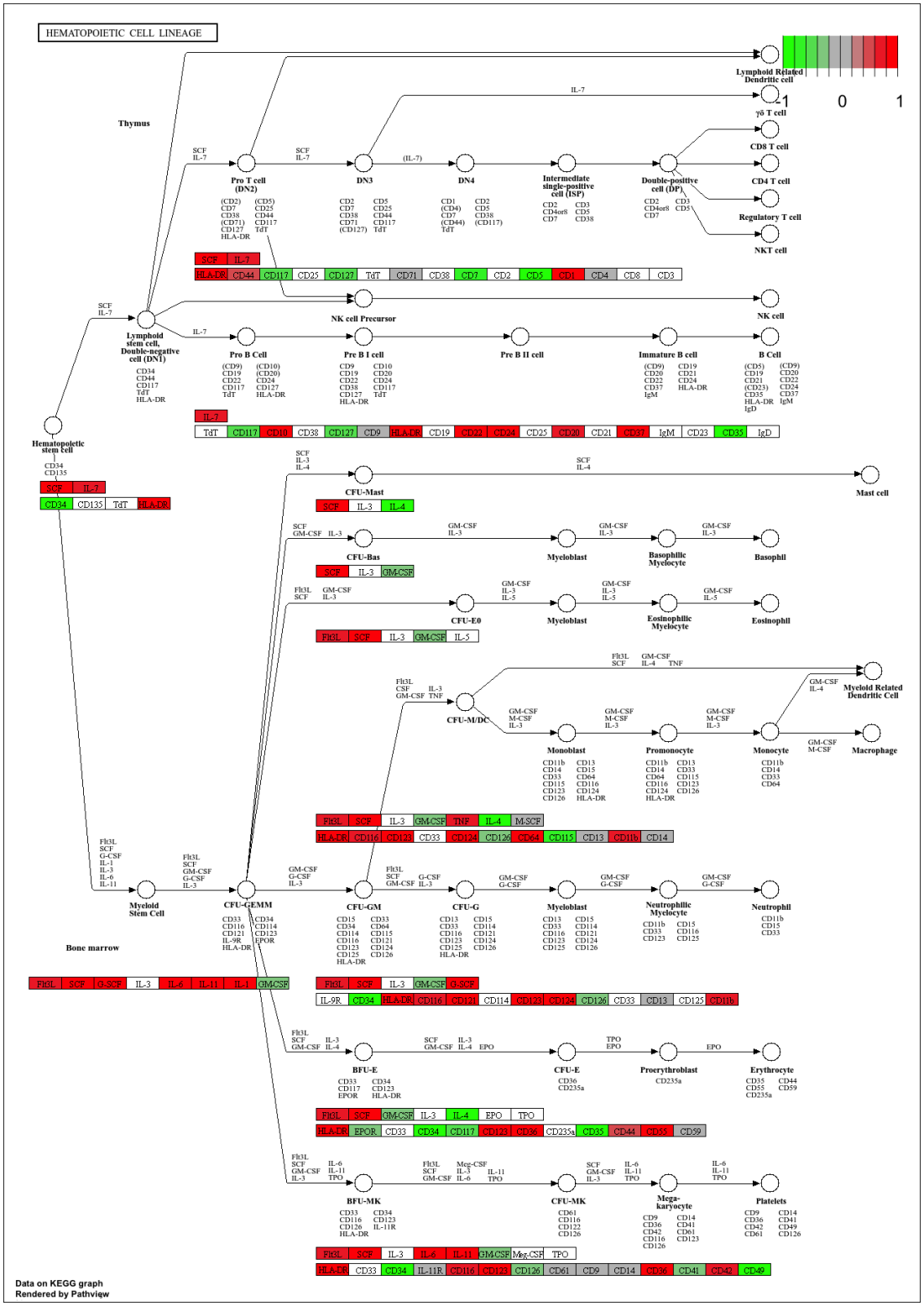
Finally, lets pass these IDs in `keggresids` to the `pathview()` function to draw plots for all the top 5 pathways.

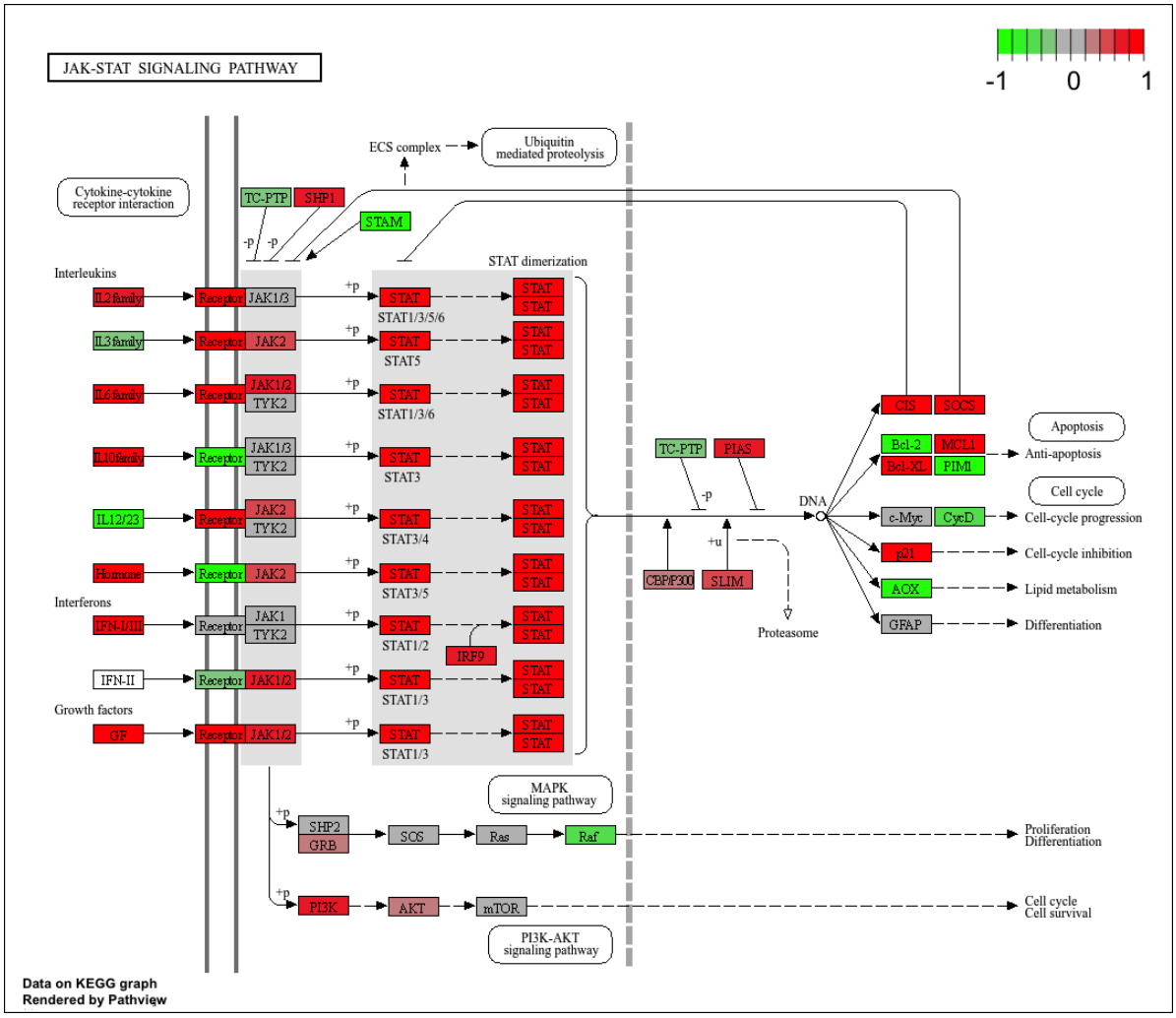
```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

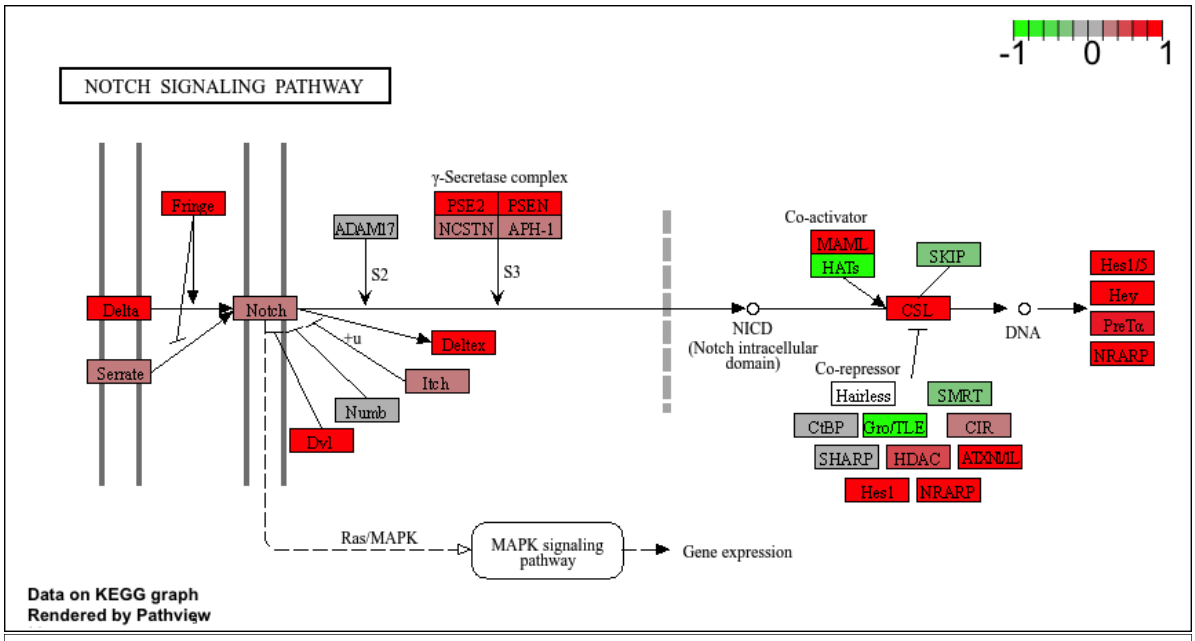
```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa00140.pathview.png
```

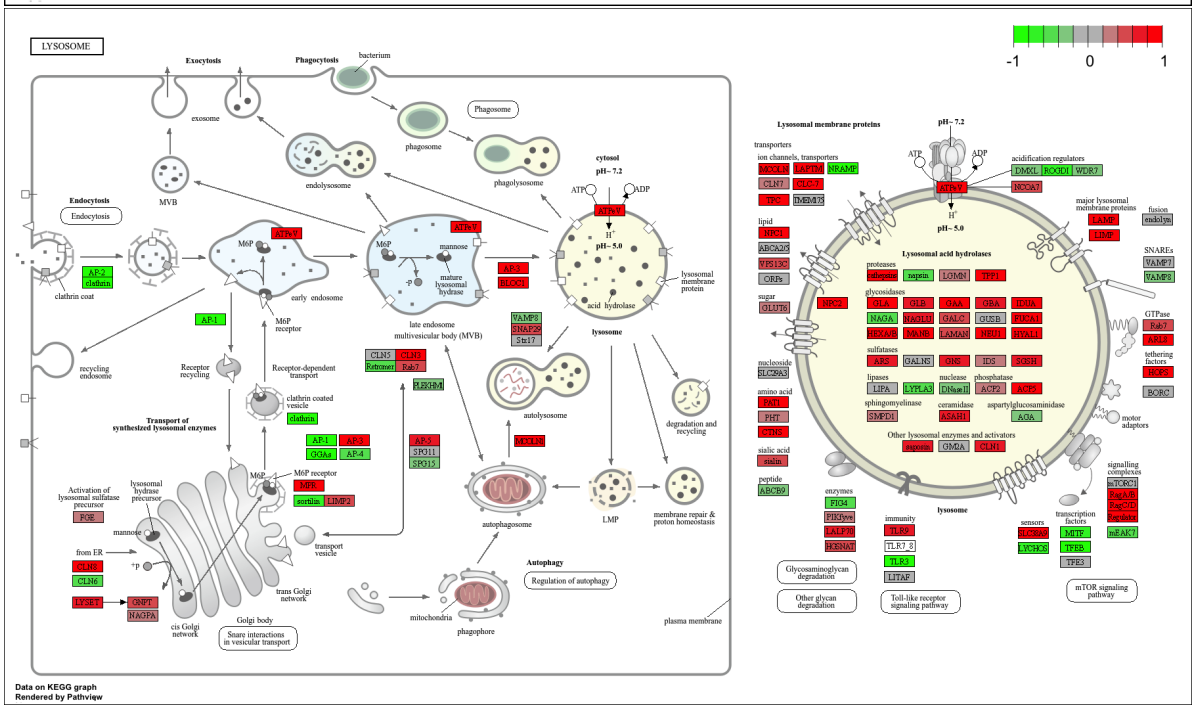
```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa04330.pathview.png
```

Here are the plots:


HEMATOPOIETIC CELL LINEAGE
Lymphoid Related Dendritic cell
γδ T cell
CD8 T cell
CD4 T cell
Regulatory T cell
NKT cell
Thymus
Pro T cell (DN2)
DN3
DN4
Intermediate single-positive cell (ISP)
Double-positive cell (DP)
NK cell Precursor
NK cell
Lymphoid stem cell, Double-negative cell (DN1)
Pro B Cell
Pre B I cell
Pre B II cell
Immature B cell
B Cell
Hematopoietic stem cell
CFU-Mast
Mast cell
CFU-Bas
Myeloblast
Basophilic Myelocyte
Basophil
CFU-E0
Eosinophilic Myelocyte
Eosinophil
CFU-M/DC
Myeloid Related Dendritic Cell
Monoblast
Promonocyte
Monocyte
Macrophage
Myeloid Stem Cell
CFU-GEMM
CFU-GM
CFU-G
Neutrophilic Myelocyte
Neutrophil
Bone marrow
BFU-E
CFU-E
Proerythroblast
Erythrocyte
BFU-MK
CFU-MK
Mega-karyocyte
Platelets
Data on KEGG graph
Rendered by Pathview


JAK-STAT SIGNALING PATHWAY
-1
0
1
ECS complex
Ubiquitin mediated proteolysis
Cytokine-cytokine receptor interaction
TC-PTP
SHP1
STAM
-p
-p
STAT dimerization
Interleukins
IL2family
IL3family
IL6family
IL10family
IL12/23
Hormone
Interferons
IFN-I/III
IFN-II
Growth factors
GF
Receptor
JAK1/3
JAK2
JAK1/2
TYK2
JAK1/3
TYK2
JAK2
TYK2
JAK2
JAK1
TYK2
JAK1/2
JAK1/2
+p
STAT
STAT1/3/5/6
STAT5
STAT1/3/6
STAT3
STAT3/4
STAT3/5
STAT1/2
STAT1/3
STAT1/3
IRF9
TC-PTP
PIAS
-p
+u
CBP/P300
SLIM
Proteasome
DNA
CIS
SOCS
Bcl-2
MCL1
Bcl-XL
PIM1
c-Myc
CycD
p21
AOX
GFAP
Apoptosis
Anti-apoptosis
Cell cycle
Cell-cycle progression
Cell-cycle inhibition
Lipid metabolism
Differentiation
MAPK signaling pathway
SHP2
GRB
SOS
Ras
Raf
Proliferation
Differentiation
PI3K
AKT
mTOR
Cell cycle
Cell survival
PI3K-AKT signaling pathway
Data on KEGG graph
Rendered by Pathview





Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

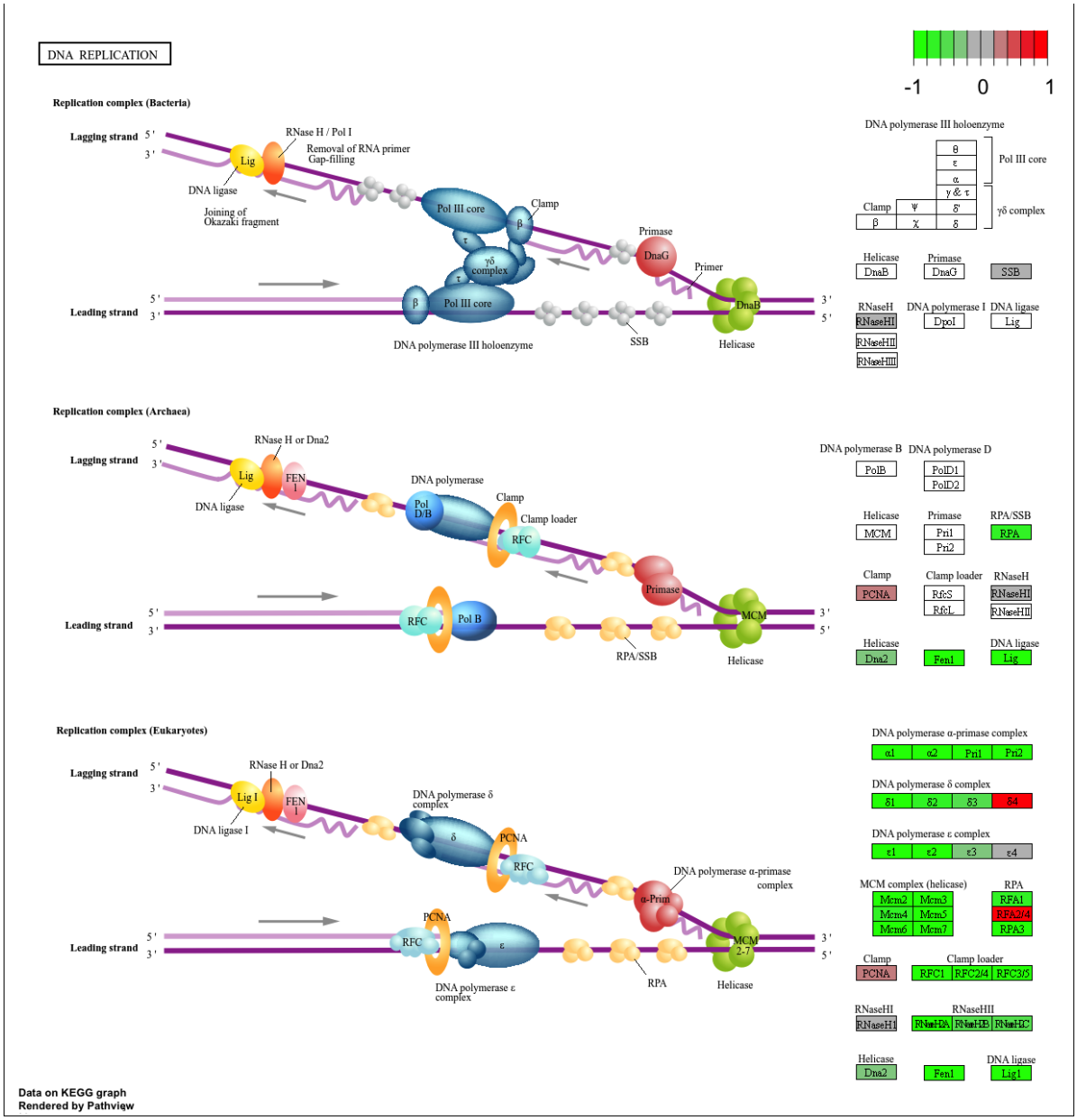
```r
keggrespathways_down <- rownames(keggres$less)[1:5]
keggresids_down <- substr(keggrespathways_down, start=1, stop=8)
keggresids_down
```

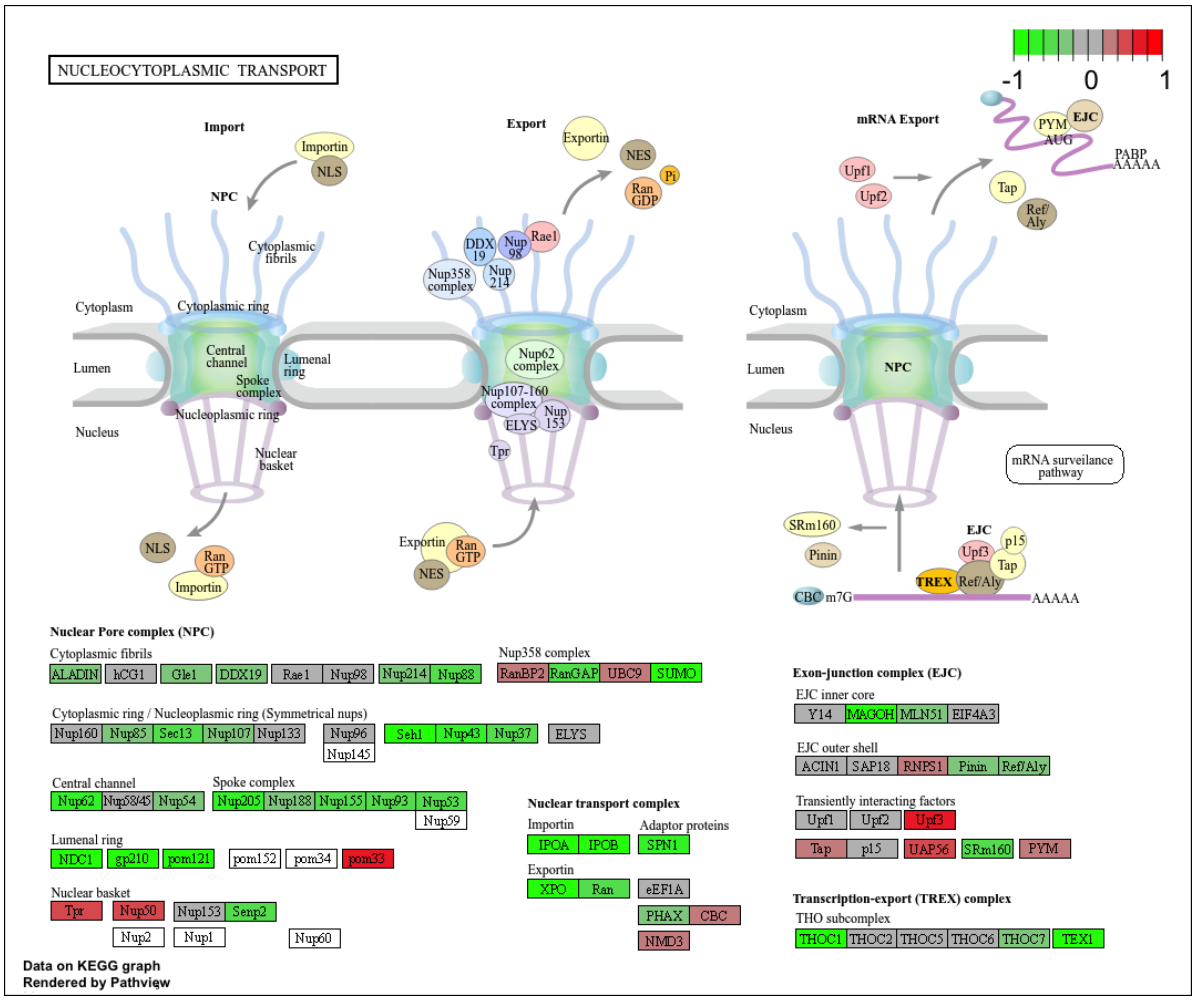
```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

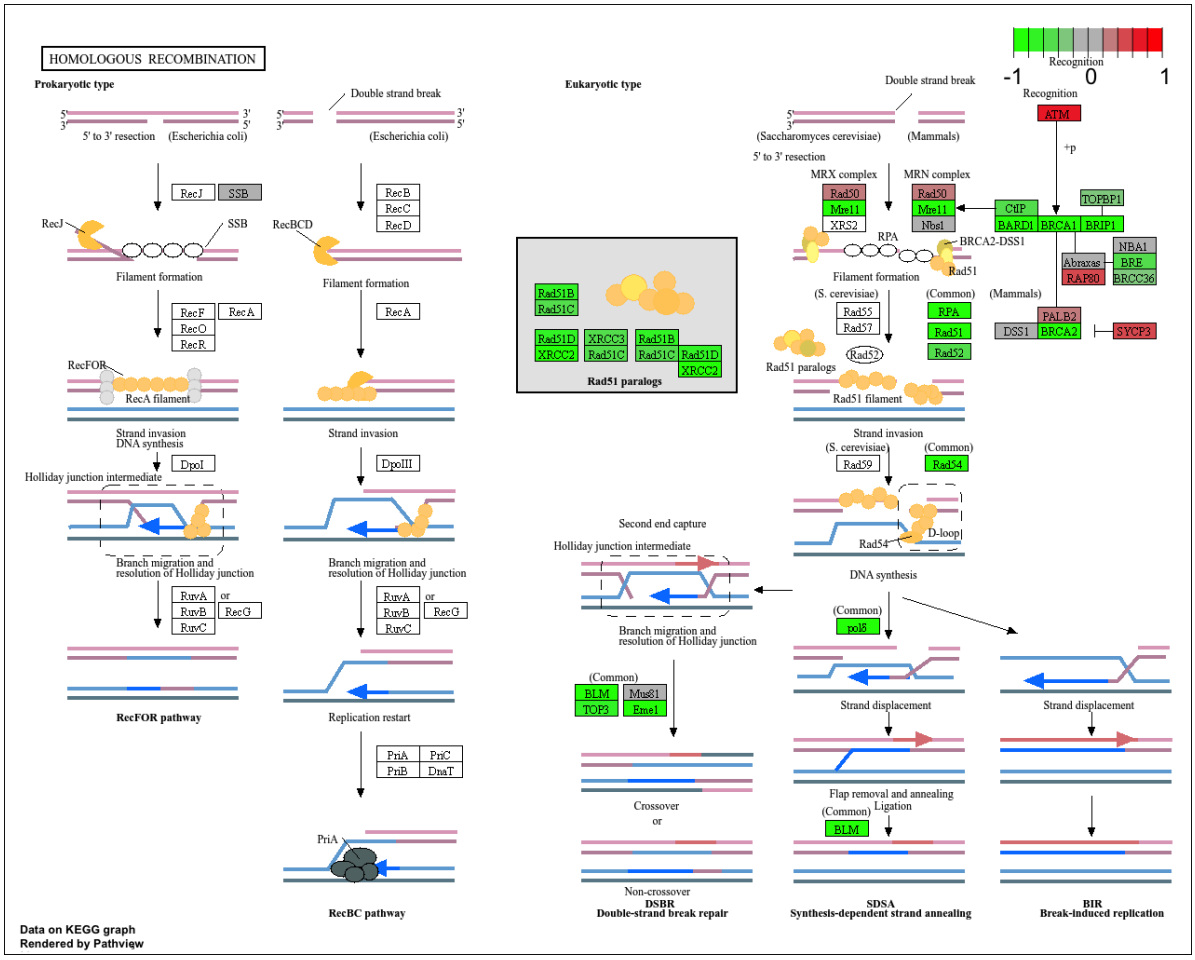
```r
pathview(gene.data = foldchanges,
         pathway.id = keggresids_down,
         species = "hsa")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/mayalsoufi/Desktop/BIMM 143/Class14
```

```
Info: Writing image file hsa04114.pathview.png
```




NUCLEOCYTOPLASMIC TRANSPORT
-1
0
1
Import
Importin
NLS
NPC
Cytoplasmic fibrils
Cytoplasm
Cytoplasmic ring
Central channel
Lumenal ring
Spoke complex
Lumen
Nucleoplasmic ring
Nucleus
Nuclear basket
NLS
Ran GTP
Importin
Export
Exportin
NES
Pi
Ran GDP
Rae1
DDX 19
Nup 98
Nup358 complex
Nup 214
Nup62 complex
Nup107-160 complex
ELYS
Nup 153
Tpr
Exportin
Ran GTP
NES
mRNA Export
EJC
PYM
AUG
PABP
AAAAA
Upf1
Upf2
Tap
Ref/ Aly
Cytoplasm
Lumen
NPC
Nucleus
mRNA surveilance pathway
SRm160
Pinin
EJC
p15
Upf3
Tap
TREX
Ref/Aly
CBC
m7G
AAAAA
Nuclear Pore complex (NPC)
Cytoplasmic fibrils
ALADIN
hCG1
Gle1
DDX19
Rae1
Nup98
Nup214
Nup88
Nup358 complex
RanBP2
RanGAP
UBC9
SUMO
Cytoplasmic ring / Nucleoplasmic ring (Symmetrical nups)
Nup160
Nup85
Sec13
Nup107
Nup133
Nup96
Nup145
Seh1
Nup43
Nup37
ELYS
Central channel
Nup62
Nup58/45
Nup54
Spoke complex
Nup205
Nup188
Nup155
Nup93
Nup53
Nup59
Lumenal ring
NDC1
gp210
pom121
pom152
pom34
pom33
Nuclear basket
Tpr
Nup50
Nup153
Senp2
Nup2
Nup1
Nup60
Nuclear transport complex
Importin
IPOA
IPOB
Adaptor proteins
SPN1
Exportin
XPO
Ran
eEF1A
PHAX
CBC
NMD3
Exon-junction complex (EJC)
EJC inner core
Y14
MAGOH
MLN51
EIF4A3
EJC outer shell
ACIN1
SAP18
RNPS1
Pinin
Ref/Aly
Transiently interacting factors
Upf1
Upf2
Upf3
Tap
p15
UAP56
SRm160
PYM
Transcription-export (TREX) complex
THO subcomplex
THOC1
THOC2
THOC5
THOC6
THOC7
TEX1
Data on KEGG graph
Rendered by Pathview



Section 3. Gene Ontology (GO)

We can also do a similar procedure with gene ontology. Similar to above, go.sets.hs has all GO terms. go.subs.hs is a named list containing indexes for the BP, CC, and MF ontologies. Let's focus on BP (a.k.a Biological Process) here.

```
data(go.sets.hs)
data(go.subs.hs)

gobpsets = go.sets.hs[go.subs.hs$BP]
gobpres = gage(foldchanges, gsets=gobpsets)
lapply(gobpres, head)
```

```
$greater
```

```
                                          p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       8.132355e-05  3.836423 8.132355e-05
GO:0002009 morphogenesis of an epithelium 1.301491e-04  3.672346 1.301491e-04
GO:0048729 tissue morphogenesis           1.324453e-04  3.663702 1.324453e-04
GO:0007610 behavior                       1.786592e-04  3.585306 1.786592e-04
GO:0035295 tube development               5.558494e-04  3.273391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  5.613074e-04  3.277405 5.613074e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1804788      113 8.132355e-05
GO:0002009 morphogenesis of an epithelium 0.1804788      339 1.301491e-04
GO:0048729 tissue morphogenesis           0.1804788      424 1.324453e-04
GO:0007610 behavior                       0.1825897      426 1.786592e-04
GO:0035295 tube development               0.3325708      391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  0.3325708      257 5.613074e-04

$less
                                            p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.735928e-15 -8.047667 1.735928e-15
GO:0000280 nuclear division              4.808128e-15 -7.923688 4.808128e-15
GO:0007067 mitosis                       4.808128e-15 -7.923688 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.312717e-14 -7.781713 1.312717e-14
GO:0007059 chromosome segregation        2.149699e-11 -6.868718 2.149699e-11
GO:0000236 mitotic prometaphase          1.794819e-10 -6.689146 1.794819e-10
                                                 q.val set.size         exp1
GO:0048285 organelle fission             6.551876e-12      376 1.735928e-15
GO:0000280 nuclear division              6.551876e-12      352 4.808128e-15
GO:0007067 mitosis                       6.551876e-12      352 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.341596e-11      362 1.312717e-14
GO:0007059 chromosome segregation        1.757594e-08      142 2.149699e-11
GO:0000236 mitotic prometaphase          1.222870e-07       84 1.794819e-10

$stats
                                          stat.mean     exp1
GO:0007156 homophilic cell adhesion        3.836423 3.836423
GO:0002009 morphogenesis of an epithelium  3.672346 3.672346
GO:0048729 tissue morphogenesis            3.663702 3.663702
GO:0007610 behavior                        3.585306 3.585306
GO:0035295 tube development                3.273391 3.273391
GO:0060562 epithelial tube morphogenesis   3.277405 3.277405
```

Section 4. Reactome Analysis

Reactome is database consisting of biological molecules and their relation to pathways and processes. Reactome, such as many other tools, has an online software available (https://reactome.org/) and R package available (https://bioconductor.org/packages/release/bioc/html/ReactomePA.html).

Let's now conduct over-representation enrichment analysis and pathway-topology analysis with Reactome using the previous list of significant genes generated from our differential expression results above.

First, using R, output the list of significant genes at the 0.05 level as a plain text file:

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Then, to perform pathway analysis online go to the Reactome website (https://reactome.org/PathwayBrowser/#TOOL=AT). Select "choose file" to upload your significant gene list. Then, select the parameters "Project to Humans", then click "Analyze".

> Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

```
head(keggres$greater)
```

```
                                       p.geomean stat.mean       p.val
hsa04640 Hematopoietic cell lineage   0.002768235  2.840168 0.002768235
hsa04630 Jak-STAT signaling pathway   0.005047208  2.596424 0.005047208
hsa00140 Steroid hormone biosynthesis 0.007156705  2.532159 0.007156705
hsa04142 Lysosome                     0.009770966  2.351292 0.009770966
hsa04330 Notch signaling pathway      0.018381844  2.120084 0.018381844
hsa04916 Melanogenesis                0.018931153  2.092166 0.018931153
                                          q.val set.size        exp1
hsa04640 Hematopoietic cell lineage   0.3840765       55 0.002768235
hsa04630 Jak-STAT signaling pathway   0.3840765      109 0.005047208
hsa00140 Steroid hormone biosynthesis 0.3840765       31 0.007156705
hsa04142 Lysosome                     0.3932814      118 0.009770966
```

```
hsa04330 Notch signaling pathway      0.4331850       46 0.018381844
hsa04916 Melanogenesis                0.4331850       90 0.018931153
```

```r
head(keggres$less)
```

```
                                        p.geomean stat.mean        p.val
hsa04110 Cell cycle                   9.366328e-06 -4.369134 9.366328e-06
hsa03030 DNA replication              9.592718e-05 -3.946505 9.592718e-05
hsa03013 RNA transport                1.305049e-03 -3.045177 1.305049e-03
hsa03440 Homologous recombination     3.106306e-03 -2.848189 3.106306e-03
hsa04114 Oocyte meiosis               3.888649e-03 -2.688749 3.888649e-03
hsa00010 Glycolysis / Gluconeogenesis 9.125398e-03 -2.398384 9.125398e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                   0.001507979      121 9.366328e-06
hsa03030 DNA replication              0.007722138       36 9.592718e-05
hsa03013 RNA transport                0.070037607      144 1.305049e-03
hsa03440 Homologous recombination     0.125028834       28 3.106306e-03
hsa04114 Oocyte meiosis               0.125214484      102 3.888649e-03
hsa00010 Glycolysis / Gluconeogenesis 0.213990959       53 9.125398e-03
```

The pathway with the most significant Entities p-value was Cell cycle (hsa04110). Several of the top pathways overlapped with those identified in the previous KEGG over-representation analysis, however the exact rankings and significance values differed. These differences arise because the enrichKEGG method uses a thresholded list of significant genes, whereas gage uses continuous fold change values for all genes. As a result, gage can detect coordinated but modest changes across a pathway, while enrichKEGG focuses only on genes passing a significance cutoff. Differences in statistical models and gene filtering may also contribute to variation between the two approaches.